Documentation Week 3 – Liv

Presentation Link
[STAT 390_Presentation 3.pptx]

Dashboard/Code Link
[Legal Menu Dashboard]

[Pre Legal Menu Dashboard]

[Code]

What I Did

This week, I built on the previous data cleaning and preparation work I led in RStudio, where I standardized the CAR dataset, formatted timestamps, replaced missing values, and added time-based variables (hour, day, and season) to support filtering in Power BI.

After Krish shared his expanded dataset containing multiple activity variables, I worked with Ella to restructure and update our data model. Together, we added Activity 4 and Activity 5 to capture deeper call paths and adjusted the hierarchy so the Pre-Legal Seniors Menu and similar multi-step flows would drill down correctly in Power BI. We verified that these new activity levels displayed properly and that the full EP → Flow → Activity chain worked across all menus.

How This Helped the Project

- Extended the activity hierarchy so Power BI now shows the complete call path.
- Ensured new menus like *Pre-Legal Seniors Menu* display correctly in the drill-down chart.
- Preserved clean labeling and consistent time variables for future analysis.

Next Steps

- Continue validating that Activities 4 and 5 populate consistently across all data sources.
- Add KPIs summarizing average activity depth per EP.
- Test additional menus to confirm the new structure supports all drill-down paths.

Summary

Earlier cleaning and restructuring work made it possible to integrate Krish's updated data smoothly. Collaborating with Ella, I focused on adding new activity fields and reconfiguring the Power BI hierarchy so multi-step menus like *Pre-Legal Seniors* now drill down accurately and align with the true call flow structure.

Documentation Week 2 - Liv

Presentation Link

[STAT 390 _ Presentation 2.pptx]

Dashboard/Code Link

[Dashboard]

[Code]

What I Did

- Focused primarily on the data cleaning and preparation in RStudio to make sure the CAR dataset could be used effectively for the Power BI hierarchy.
- Wrote the R scripts to:
  - Add a submenu count column (submenu_count) for each call session.
  - Replace NA values with "Unknown" labels for EP, Flow, and Activity levels so they would appear properly in Power BI.
  - Convert ep_name, flow_name, and activity_name variables to character type to prevent data-type conflicts.
- Investigated why EP names were showing up as NA in Power BI, discovered that the values were missing on some rows within the same contact_session_id, and wrote the code to fix this by filling the values in R (propagating EP names down within each call).
- Extended the same logic to fill missing Flow names while ensuring that the first step of each call remained "Unknown Flow" if that was how it was logged.
- Verified the fixes by checking a sample of contact_session_ids to confirm that all EPs were now filled correctly and that Flows matched the expected call order.
- Re-exported the cleaned dataset as a CSV for use in Power BI.
- In Power BI, helped configure the hierarchical stacked column chart and tested the drill-down functionality, confirming that clicking an EP revealed its related Flows and Activities.
- Worked jointly with Ella on final formatting and slicers but took the lead on data transformation, debugging, and validation in R.

How This Helped the Project

- Enabled accurate hierarchical visualization of caller navigation through EPs, Flows, and Activities.
- Reduced missing or misaligned data that previously caused (blank) categories in Power BI.

- Created a clean data structure that reflects the true phone menu hierarchy, allowing meaningful drill-down analysis.
- Supported more reliable measurement of submenu depth and caller navigation efficiency.

Columns Used

- contact_session_id — unique call identifier
- ep_name — main entry point (top-level menu)
- flow_name — submenu within EP
- activity_name — lowest-level menu interaction
- activity_start_timestamp — used to order menu steps
- day_type — weekday vs. weekend indicator
- hours_status — open vs. closed hours indicator
- submenu_count — number of submenus navigated per call

Insights Derived

- Certain menus such as Legal Menu and Pre-Legal Menu generated the highest submenu counts, suggesting these areas may be overly complex or confusing for callers.
- Queue-related menus (e.g., “Queues” and “ClosedQueueMenu”) frequently appeared as standalone EPs, signaling potential bottlenecks or repeated redirections.
- The hierarchy revealed distinct patterns in caller flow, allowing for potential menu simplification and rerouting opportunities to improve efficiency.

Questions for Cynthia

1. Should our Y-axis metric (distinct call count) be normalized by date or time to better account for call volume variation?
2. Would including average submenu count by EP strengthen the analysis of menu efficiency?
3. Is it worth adding call duration or termination reason to the hierarchy to link complexity with call outcomes?
4. Should “Unknown” categories remain visible for transparency, or be filtered out for cleaner visuals?

Issues Faced & Resolutions

Issue	Cause	Fix
EP names showing as NA	EP only recorded on one row per call; missing elsewhere in the session	Grouped by contact_session_id and used fill() in R to propagate EP names down
Flow names missing at start of call	Flow not logged in first menu step	Kept first “Unknown Flow” but filled subsequent steps downward
Hierarchy drill-down showing blanks	Missing or mismatched data labels	Replaced all NAs with “Unknown” placeholders
Legend not fully visible in chart	Too many long category labels	Reformatted labels and adjusted legend layout
Metric dependent on time window	Original Y-axis based on call duration	Switched to distinct count of contact_session_id for stability

Next Steps

- Add all the data into RStudio to show combining and add updated months there
- Add slicers for date, day_type, and hours_status in Power BI to allow dynamic filtering.
- Incorporate a card visual showing total distinct calls and average submenu count per EP.
- Explore menu efficiency metrics such as average submenus per EP or flow.
- Group related EPs (e.g., all “Legal” menus) for higher-level trend analysis.
- Review findings with Cynthia to connect insights to potential operational recommendations.

Summary

Through detailed data cleaning and restructuring in R, I prepared the CAR dataset for Power BI’s hierarchical visualization. The resulting stacked column chart with drill-down now clearly shows how callers move from main menus to submenus and activities, revealing points of complexity and inefficiency. This work lays the foundation for further exploration of call routing patterns and menu optimization to improve client experience and reduce navigation time.

Documentation Week 1 - Liv

Presentation Link

STAT 390: Presentation 1

Dashboard/Code Link

Dashboard

CAR Dashboard

What I Did

Our team is developing a Power BI dashboard to analyze Legal Aid's phone system and identify inefficiencies in how clients navigate automated menus. The goal is to reduce caller frustration and support data-driven decisions for simplifying call flows.

My key contributions included:

- New Metric Ideation: Helped design and define Submenu Efficiency (average number of steps before failure).
    - Currently, this variable measures the number of menus navigated per call.
- Power BI Support: Helped group members import data and start exploratory visuals.
- Data Cleaning/EDA (All Calls Dataset):
    - Edwin and I worked on the following:
    - Converted the Called number column to Text, since it contained special characters (#) that caused Power BI import errors.
    - Removed all records with Duration = 0 to prevent skewed averages.
    - Standardized Contact Session ID as a consistent session identifier, since External Caller ID varied between calls.
- Exploratory Analysis (CAR Dataset):
    - I worked on the following on my own:
    - Created a variable to count how many menus are navigated.
        - Using COUNTIF formula in Excel
    - Clustered Column Charts:
        - Count of Calls by Average Number of Submenus Navigated
        - Average Number of Submenus Navigated by Activity Type

How This Helped the Project

- Columns Used:
    - From All Calls Dataset: Duration, Redirect reason, Call outcome, User type, Answer indicator.
    - From CAR Dataset: Contact Session ID, Activity Name, Menus Navigated (new).

- Insights Derived:
    - “Deflection” and “Unconditional” redirect reasons have the longest average durations, suggesting possible routing inefficiencies.
    - “FollowMe” and “NoAnswer” are most frequent redirect reasons but have shorter durations, indicating quick hangups or missed calls.
    - Automated systems (like VirtualLine and AutomatedAttendantVideo) tend to have shorter call times, likely due to quick routing or early disconnections.
    - These findings helped guide metric design (Submenu Efficiency and Redirection Performance) which quantify caller effort and system success.
    - A better understanding of the average amount of submenus navigated.
    - Which activities are creating the most complication in having to navigate many submenus.
- Questions for Cynthia:
    - In the Redirect Reason column, what does an “N/A” value represent?
        - Does it mean the call was never redirected, or that the redirect data wasn’t captured for that session?
    - When a call ends in a ClosedQueueMenu, does Webex record that as a “failed” or “closed” outcome in the data, or is that outcome only visible in the CAR dataset?
    - For Duration, does that value reflect the entire call length (including ringing and menu navigation), or only the time once the call is answered or queued?

Issues Faced & Resolutions

Issue	Cause	Fix
Data Type Errors	Called number contained symbols like #, breaking Power BI import	Converted to Text in Power Query
Zero-Duration Calls	Invalid data rows	Filtered out during import
Unstable Caller IDs	External Caller ID changed per call	Used Contact Session ID for unique session tracking

Power BI Web Limitations	Couldn't change column data types or apply transformations online	Switched to Power BI Desktop (on Edwin's PC) for full functionality

Next Steps

- Expand EDA using both All Calls and CAR datasets to analyze:
    - Call outcomes, duration distributions, and submenu usage.
- Refine metrics:
    - Submenu Efficiency – average steps before a failed call.
        - Create a graph showing average submenus navigated to and if there are any connections between the amount of submenus navigated to with other variables.
        - Make the variable more complex by adding filters for EP Name, Flow Name, and Activity Name.
    - Redirection Performance – % of redirected calls successfully connected.
- Add new metrics:
    - Closed Queue Exposure Rate – % of calls ending in Closed Queue.
    - Navigation Time Wasted – avg. time spent before reaching a dead end.
- Build final dashboard model with improved filters and visual interactivity.

Summary

This phase focused on cleaning data, exploring redirection trends, and defining new performance metrics to measure system efficiency. This was mostly a more basic dive into the data and figuring out Power BI, but the findings provide a foundation for deeper analysis in Presentation 2, by connecting caller navigation patterns to specific bottlenecks in Legal Aid's phone system.